Module 6
Project Data Cleaning

Student: Yu Lin **Project:** Community Safety Resource Planner
Prepared: October 4, 2026 **Analysis:** January 1, 2025 onward

Project vision and business question

A City of Los Angeles street-lighting program manager needs a defensible way to prioritize locations for on-site lighting assessments. From January 1, 2025 onward, how are nighttime reported-offense records distributed across LAPD areas, and which patterns warrant further assessment?

The main measures are nighttime offense-record count, distinct nighttime CaseNo count, and nighttime share of eligible records. Nighttime means 18:00-05:59, an individual analytical choice. These are reported-event measures; they do not establish population crime rates, lighting deficiencies, or a causal effect of lighting.

Fixed team snapshot and preservation

Item	Value
Team file	LAPD_NIBRS_Offenses_Dataset_20261004.csv
Active input	Project/LAPD_NIBRS_Offenses_Dataset.csv (byte-identical copy)
Dimensions	489,955 rows x 31 original columns; 489,955 rows x 44 cleaned columns
Version context	Team file pinned October 4; team described the source as updated September 29, 2026. Upstream metadata time was not independently reverified.
Observed coverage	1800-01-06 through 2026-09-19; this is observed coverage, not proof of complete reporting.

Raw SHA-256, unchanged before/after processing:

`97a8dbadafe60eee4b4d86e99769df3aaa23307d45539283879414f8760b39e5`

GitHub submission details

Existing personal branch: anselyulin
Branch URL: https://github.com/zayra-rwx/community-safety-resource-planner/tree/anselyulin
Submitted analysis commit ID: 099cb874e738e2d0f8fccf31c74aa55223e3b4f1
Instructor repository access: Not independently confirmed.

The submitted analysis commit contains the scripts, executed notebook, sample, dependencies and verification evidence. A subsequent documentation commit adds this PDF and records the analysis commit ID in the READMEs. The analysis files and results are unchanged in that follow-up.

Local verification	Result
Full-data checks	13 / 13 passed
Individual record checks	5 / 5 matched
Separate rerun	Cleaned CSV and both area-metric CSVs were byte-identical

14-item checklist | Inspect and clean

Status labels follow the assignment. Evidence paths are relative to Project/module5_cleaning unless otherwise stated. Full results and decision rules appear on pages 4-9.

1. State the project vision and question [Done]

The vision, decision-maker, question and three measures are on page 1. Use occurrence dates from 2025-01-01 onward, consistent with the team scope; the fixed file currently extends to 2026-09-19. See clean_nibrs.py and page 6.

2. Define one row [Unresolved]

One row is provisionally treated as one reported-offense record. UniqueNIBRNo is unique and nonblank for all 489,955 rows; CaseNo repeats. The official semantic row definition remains unconfirmed, so offense records and distinct cases are reported separately. Do not interpret rows as unique incidents or victims. Evidence: full_validation.json; page 5.

3. Preserve the source [Done]

The exact team filename, adoption date, observed coverage and SHA-256 are on page 1. The raw file was never edited by cleaning. The original desktop team file and a previous-snapshot backup remain available. Load the team snapshot as strings; write only derived outputs. Reproduction instructions are on page 9.

4. Inspect columns and types [Done]

Read all columns as strings to protect IDs. Parse Date Rptd and Date OCC; validate HHMM and derive an integer hour; convert coordinates temporarily to numeric for checks. Preserve original text columns and add derived columns. Nonblank date/time/coordinate conversion failures are zero. Missing Date Rptd and coordinates remain missing. See page 5.

5. Standardize text [Done]

Trim surrounding whitespace without changing case, internal text or code meanings. AREA NAME changes on all 489,955 rows. For example, Central plus 13 trailing spaces becomes Central; AREA "01" remains "01". Empty-after-trim values remain missing. True raw values and actual cleaned text are shown on page 7.

6. Investigate duplicates [Done]

Exact duplicate rows: 0 before and after. Duplicate UniqueNIBRNo values: 0. There are 44,390 repeated CaseNo values spanning 94,224 rows. Keep distinct offense IDs sharing a case; do not drop those rows. See pages 5 and 7.

7. Handle missing data [Done]

Affected-column missing counts before and after are listed on page 5. No values are imputed and no rows are dropped. A blank weapon field does not mean no weapon. Missing coordinates prevent mapping, not area-level counting; missing dates/times or area keys would exclude records from the relevant metric denominator.

14-item checklist | Validate results

8. Check suspicious values [Done]

Flag and retain 4 records with occurrence year <= 1900, 550 double-zero coordinate records, and 131 coordinate-missing records. Midnight is valid but ambiguous, so retain it and compute a sensitivity comparison. See pages 5-8.

9. Verify merges [Not applicable]

No external table was merged in this cleaning assignment. Therefore merge keys, unmatched-key counts and join-induced extra rows are not applicable. Any later streetlight join will need its own key/cardinality and row-count checks.

10. Reconcile changes [Done]

489,955 original rows become 489,955 cleaned rows: 0 added, 0 deleted. The identifier set is unchanged. The 13 new columns hold parsed values/flags. Analysis filtering retains 380,091 eligible rows and excludes 109,864 earlier records; excluded rows still remain in the cleaned CSV. See page 6.

11. Check calculations [Done]

For the full period, nighttime share is 179,209 / 380,091 = 47.15%. Excluding 0000, it is 168,879 / 369,761 = 45.67%. Missing area keys are excluded, and distinct-case counts ignore missing CaseNo; neither is missing in this snapshot. Each area's numerator, denominator and case count are on page 6. No mean of missing-filled values is used.

12. Verify individual records [Done]

Five difficult records cover whitespace/leading zeros, repeated cases, midnight, an early date, and missing coordinates. In the current verification pass, expectations and reasons were saved from the raw data to outputs/record_expectations.json before the validator read the cleaned CSV. All five matched. See original / expected / actual evidence on pages 7-8.

13. Test reproducibility [Done]

All four notebook code cells ran in order in a fresh Python kernel using Python 3.14.7 and pandas 3.0.3; all full-data checks passed. A separate Python process reran cleaning and produced byte-identical cleaned and area-metric files. Production-code tests checked 2024 exclusion, 2025/2026 inclusion, missing area keys and midnight denominators. See page 9.

14. Document remaining limitations [Unresolved]

The limitations are documented, but row semantics, special code meanings, historical dates, midnight meaning and source completeness remain unresolved. These affect interpretation, period comparability, mapping and lighting relevance. The latest observed occurrence is 2026-09-19; the file label is not an assertion of coverage through October 4. See page 9.

Check scope

The 13 automated checks are not the same thing as the 14 assignment checklist items. The checklist also includes interpretation, documentation, and the no-merge explanation. All original columns and all 13 derived columns are compared against raw-derived expected values by UniqueNIBRNo; documented whitespace/blank normalization is allowed.

Cleaning decision log

The rules below were documented before the current scope rerun. The raw source is retained intact. Exceptions remain visible rather than being silently filled or deleted.

Rule	Reason	Check / outcome
Read all source columns as strings	Preserve IDs and code-leading zeros	UniqueNIBRNo set unchanged; code fields match source text after trim.
Strip outer whitespace; empty strings become missing	Remove padding without erasing meaningful text	AREA NAME changed on 489,955 rows; original/expected/actual example on page 7.
Keep weapon blanks missing	Blank does not prove no weapon	345,971 blanks in each weapon field before and after.
Parse dates; preserve source text	Enable date filtering and retain provenance	0 nonblank date parse failures; missing reported dates remain missing.
Validate four-digit HHMM; derive hour	Use a consistent time rule	0 invalid time values; leading zeros retained.
Keep 0000; flag midnight	The source may use midnight literally or as an unknown time	Recompute both numerator and denominator without 0000; page 6.
Flag occurrence year <= 1900; do not rewrite	Unusual dates require review, not guessed corrections	4 flagged; all retained in cleaned data and outside analysis period.
Check coordinates numerically; retain original text	Map only usable coordinates while retaining area evidence	131 missing, 550 double-zero, 0 outside global ranges.
Do not deduplicate CaseNo	A case can have multiple distinct offense IDs	440,121 distinct cases; repeated-case records retained; page 7.
Use Date OCC >= 2025-01-01, with no 2026 cutoff	Align with the team's stated January 1, 2025 onward scope	Independent parsed-year >= 2025 comparison; production boundary test passed.
Require nonmissing AREA and valid date/time for metrics	Give each share an explicit eligible denominator	All 380,091 in-period rows satisfy those conditions; page 6.
No external merge and no causal claim	Lighting conditions and population denominators are absent	No join or row expansion; describe assessment priorities only.
Compare by UniqueNIBRNo and protect raw file	Detect undocumented changes and preserve identity	0 field mismatches; raw hash unchanged; separate rerun hashes match.

Rule revisions

September 27: earlier work corrected a December 31 cutoff error and the midnight-sensitivity denominator.
October 4: adopted the team CSV, then removed the 2026 upper cutoff after checking Discord and receiving user confirmation. The earlier 2025-only results are archived. The two changes are distinct: source version and analysis scope.

Full-data evidence | Types and missingness

Field / conversion	Failed nonblank values	Treatment
Date Rptd -> parsed timestamp	0	7,787 missing values stay missing
Date OCC -> parsed date	0	Preserve raw timestamp text; parsed output stores date
Time OCC -> validated HHMM / hour	0	Keep four-digit text; derive integer hour
LAT / LON -> numeric checks	0 / 0	Preserve original text; derive mapping flags

Missing values in original columns

Column	Before	After	Decision
Date Rptd	7,787	7,787	Retain; no imputation
Group	15,480	15,480	Retain; no imputation
NIBR Code	1,149	1,149	Retain; no imputation
Crime Against	2,297	2,297	Retain; no imputation
Premise Cd	22,853	22,853	Retain; no imputation
Premise Desc	109	109	Retain; no imputation
Weapon Used Cd	345,971	345,971	Retain missing; do not infer no weapon
Weapon Desc	345,971	345,971	Retain missing; do not infer no weapon
Status	1,044	1,044	Retain; no imputation
Status Desc	1,042	1,042	Retain; no imputation
LAT	131	131	Retain; exclude from mapping only
LON	131	131	Retain; exclude from mapping only

All other original columns have zero missing values in this snapshot. Derived parsed/flag columns are documented separately; derived flags do not fill the original missing data. Missing numeric coordinates are distinguished from the literal pair (0, 0).

Duplicate and suspicious-record checks

Exact duplicates before / after: **0 / 0**. Repeated UniqueNIBRNo: **0**. Distinct CaseNo: **440,121**; repeated CaseNo values: **44,390**, covering 94,224 rows (49,834 rows beyond the first per case). All were retained.

Entire source: **15,333** midnight records; **4** occurrence years <= 1900; **131** coordinate-missing records; **550** double-zero coordinate records; **0** outside global latitude/longitude ranges. These flags may overlap.

Reconcile rows and verify calculations

Cleaning: 489,955 input rows - 0 dropped + 0 added = 489,955 cleaned rows.
Analysis: 489,955 - 109,864 records before 2025 = **380,091** eligible records. There are no missing occurrence dates, invalid times or missing area keys within the selected period. The cleaned CSV still retains the excluded historical records.

The analysis contains **223,307** records from 2025 and **156,784** from the observed portion of 2026. This is not a like-for-like comparison of two complete years.

Metric	Including 0000	Excluding 0000
Eligible offense records	380,091	369,761
Nighttime offense records	179,209	168,879
Nighttime share	47.1490%	45.6725%

Sensitivity removes the same **10,330** midnight records from both parts: 380,091 - 10,330 = 369,761 and 179,209 - 10,330 = 168,879. Global distinct nighttime CaseNo count is **161,593**. Area-specific distinct-case counts should not be summed as a global unique count.

Area results and denominators (including 0000)

AREA	Eligible records	Night records	Distinct night cases	Night share
01	31,647	14,249	13,086	45.02%
02	21,086	9,342	8,544	44.30%
03	21,881	9,789	8,799	44.74%
04	13,755	6,647	6,064	48.32%
05	14,578	6,745	5,993	46.27%
06	20,009	10,113	9,314	50.54%
07	16,472	7,877	7,071	47.82%
08	13,558	5,878	5,481	43.35%
09	17,297	7,994	7,098	46.22%
10	16,226	7,791	6,809	48.02%
11	16,563	7,884	7,181	47.60%
12	21,032	10,202	9,239	48.51%
13	18,183	8,951	8,291	49.23%
14	19,777	9,400	8,696	47.53%
15	20,623	9,617	8,306	46.63%
16	11,602	5,722	5,091	49.32%
17	15,561	6,992	6,369	44.93%
18	17,069	8,165	7,198	47.84%
19	16,073	7,731	7,012	48.10%
20	20,835	10,724	9,666	51.47%
21	16,264	7,396	6,286	45.47%

For each area, share = night records / eligible records. Counts use rows, not sums of TotalOffenseCount or TotalVictimCount. Blank group keys are excluded; a missing CaseNo would not contribute to nunique. No missing value is replaced by zero to calculate an average.

Five record checks | 1-3

Expectations were prepared from original records and documented rules before this validation pass read the cleaned CSV. The prior source text is preserved exactly; blanks and trailing spaces are explicitly identified below. The check results are reproducible in record_validation.csv.

Check 1 | leading-zero AREA code and trailing whitespace

UniqueNIBRNo: 25088246_13A_0 **Result:** Match

Field	Original	Expected	Actual
AREA NAME	Harbor + 14 trailing spaces	Harbor	Harbor
AREA	"05"	"05"	"05"

Reason: Strip only surrounding whitespace from AREA NAME; keep AREA as a two-character code with its leading zero.

Check 2 | same CaseNo with multiple offense records

UniqueNIBRNo: 25088246_290_2 **Result:** Match

Original: CaseNo 25088246 has these distinct offense IDs: 25088246_13A_0, 25088246_23F_1, 25088246_290_2.

Check	Expected	Actual
Retained offense IDs	25088246_13A_0 25088246_23F_1 25088246_290_2	25088246_13A_0 25088246_23F_1 25088246_290_2

Reason: Keep every distinct UniqueNIBRNo sharing the CaseNo; a repeated case is not automatically a duplicate offense.

Check 3 | midnight offense record

UniqueNIBRNo: 26172245_220_0 **Result:** Match

Field / rule	Original	Expected	Actual
Time OCC	0000	0000	0000
Midnight flag	Raw HHMM is 0000	True	True
Period eligible	2026 Aug 31 12:00:00 AM	True	True

Reason: Keep 0000 as a valid time, flag midnight, and exclude it only from the sensitivity numerator and denominator.

Five record checks | 4-5

Check 4 | early occurrence year

UniqueNIBRNo: 26003810_240_0 **Result:** Match

Field / rule	Original	Expected	Actual
Date OCC	1800 Jan 06 12:00:00 AM	1800 Jan 06 12:00:00 AM	1800 Jan 06 12:00:00 AM
Year <= 1900 flag	Unusual occurrence date	True	True
Period eligible	Before 2025-01-01	False	False

Reason: Retain the unusual original occurrence date, flag year <= 1900, and exclude it from the 2025 onward analysis.

Check 5 | missing coordinates

UniqueNIBRNo: 24100353_240_0 **Result:** Match

Field / rule	Original	Expected	Actual
LAT / LON	(empty) / (empty)	Missing / missing	Missing / missing
Coordinates Missing	Coordinates blank	True	True
Valid for mapping	No usable coordinates	False	False
Period eligible	2024 Dec 21 05:00:00 PM	False	False

Reason: Keep empty coordinates missing and flag them as unsuitable for mapping; keep the record for eligible area metrics.

How the shareable sample was selected

outputs/sample_50.csv contains exactly 50 cleaned records and all 44 cleaned columns. Six difficult-case IDs were selected first: the five cases above plus a double-zero-coordinate case. The other 44 IDs were sampled from the remaining cleaned data with random_state=576. The resulting file retains source order; selected difficult cases need not appear first. This sample supports inspection, not full-population verification.

One Codex suggestion and how it was checked

Codex assisted with implementing the date filter and validation. The intended rule was written as Date OCC >= 2025-01-01 after the team scope was confirmed. The cleaner compares parsed timestamps to that start date. The independent validator instead checks parsed year >= 2025 and compares the result by UniqueNIBRNo across the full dataset.

The production cleaning function was also tested on records at 2024-12-31 23:59:59, 2025-01-01 00:00:00 and 2026-01-01 00:00:00: the expected eligibility was False, True, True. A missing area key was excluded. A separate raw-data count agreed with the current denominator and nighttime totals. This catches the earlier mistake of restricting the analysis to 2025 alone.

Reproducibility, limitations and sources

Run from the repository root (local Project directory)

```
python3 -m pip install -r module5_cleaning/requirements.txt

python3 module5_cleaning/clean_nibrs.py

python3 module5_cleaning/validate_nibrs.py
```

Place the exact team snapshot at Project/LAPD_NIBRS_Offenses_Dataset.csv first and verify the SHA-256 on page 1. Use Python 3.14.7, pandas 3.0.3 and ipykernel 7.3.0. Open module5_cleaning/module5_cleaning.ipynb from its folder, restart the kernel, run all four code cells in order, and save with outputs visible. The historical module5 folder/filename is retained for compatibility; this work is Module 6.

Outputs and actual rerun evidence

The commands recreate cleaned_nibrs.csv, metrics_by_area.csv, metrics_by_area_excluding_midnight.csv, cleaning_run_metadata.json, full_validation.json, record_expectations.json, record_validation.csv and sample_50.csv under outputs/. The validator returns a nonzero exit code if a required check fails.

The executed notebook and a separate command-line process produced matching SHA-256 values for the three main output files. outputs/rerun_comparison.json records the hashes and test outcomes. Runtime/path metadata are not claimed to be byte-identical. Recreate the second run with:

```
python3 module5_cleaning/clean_nibrs.py --output \
  module5_cleaning/reruns/check/cleaned_nibrs.csv
```

Compare that cleaned CSV and the two generated metric CSVs to outputs/. rerun_comparison.json is a recorded audit artifact rather than an automatic output of the normal three-command workflow. All 13 full-data checks passed; the raw source hash remained unchanged.

Remaining limitations and their effects

Row semantics and code meanings: an official dictionary is needed before treating records as incidents, victims, or specific intervention needs.
Reporting and period completeness: reported offenses depend on reporting practices; 2026 is partial, and the observed latest date is September 19. Totals do not establish temporal comparability or full coverage.
Time and location: 0000 may be a true midnight time or a source convention. Coordinates are only screened for missingness, (0, 0), and global ranges; remaining points are not guaranteed to be correctly located in Los Angeles.
Lighting relevance: no streetlight condition, population/exposure or causal-design data were merged. A high count supports further assessment, not a finding that poor lighting caused offenses.

Source and evidence references

Dataset: team-provided LAPD CSV, exact version identified on page 1. Previously recorded official source page: LAPD NIBRS Offenses Dataset catalog.
Team scope: Discord DSO 576 Python Group, #general, October 4 at 19:20 (edited 19:21); dataset pinned in #notes-resources at 19:17, America/Los_Angeles.
Assignment: Module_6_Homework.pdf, pages 1-3.
Local evidence: README.md, cleaning_decisions.md, the two Python scripts, executed notebook and outputs/ files named above.
Submission: use the existing anselyulin branch; the submitted analysis commit is recorded on page 1. Instructor access remains unconfirmed; Gradescope submission is separate.